EMBEDDING COUNT EQUALS DEGREE EXACTLY WHEN EACH STEP OF THE GENERATING TOWER IS SEPARABLE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The reordering trick in the last paragraph uses that the strict-inequality and equality parts of this very lemma have already been proved within the counted span; gamma_2, ..., gamma_m are filled by alpha_1, ..., alpha_n (source lines 1589-1592).

Editorial note. The label lemma-separably-generated-separable names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Context span 1485-1496: the Situation (generating tower, minimal polynomials P_i, fixed algebraic closure)
- Same chapter above the span: lemma-count-embeddings-explicitly (|Mor_F(K, F-bar)| = product deg_s(P_i), source lines 1542-1554), lemma-degree-minimal-polynomial (deg(P_i) = [K_i : K_{i-1}]), lemma-multiplicativity-degrees (degrees multiply in towers)

Source reference. Original reference 09H8: sources/stacks-fields/pinned-original-source.tex, source lines 1543–1543. Prerequisite pointer invoked at 1581; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09GN: sources/stacks-fields/pinned-original-source.tex, source lines 1087–1087. Prerequisite pointer invoked at 1575; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09G9: sources/stacks-fields/pinned-original-source.tex, source lines 759–759. Prerequisite pointer invoked at 1576; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Situation 0.1. Here F be a field and K/F is a finite extension generated by elements $\alpha_1, \ldots, \alpha_n \in K$. We set $K_0 = F$ and

$$K_i = F(\alpha_1, \ldots, \alpha_i)$$

to obtain a tower of finite extensions $K = K_n/K_{n-1}/\ldots/K_0 = F$. Denote P_i the minimal polynomial of α_i over K_{i-1}. Finally, we fix an algebraic closure $\overline{F}$ of F.

Date: stacks-project:fields.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Assumptions and notation as in Situation 0.1. If each P_i is separable, i.e., each α_i is separable over K_{i-1}, then*

$$|\operatorname{Mor}_F(K, \overline{F})| = [K : F]$$

and the field extension K/F is separable. If one of the α_i is not separable over K_{i-1}, then $|\operatorname{Mor}_F(K, \overline{F})| < [K : F]$.

Proof. If α_i is separable over K_{i-1} then $\deg_s(P_i) = \deg(P_i) = [K_i : K_{i-1}]$ (last equality by Lemma 09GN). By multiplicativity (Lemma 09G9) we have

$$[K : F] = \prod [K_i : K_{i-1}] = \prod \deg(P_i) = \prod \deg_s(P_i) = |\operatorname{Mor}_F(K, \overline{F})|$$

where the last equality is Lemma 09H8. By the exact same argument we get the strict inequality $|\operatorname{Mor}_F(K, \overline{F})| < [K : F]$ if one of the α_i is not separable over K_{i-1}.

Finally, assume again that each α_i is separable over K_{i-1}. We will show K/F is separable. Let $\gamma = \gamma_1 \in K$ be arbitrary. Then we can find additional elements $\gamma_2, \ldots, \gamma_m$ such that $K = F(\gamma_1, \ldots, \gamma_m)$ (for example we could take $\gamma_2 = \alpha_1, \ldots, \gamma_{n+1} = \alpha_n$). Then we see by the last part of the lemma (already proven above) that if γ is not separable over F we would have the strict inequality $|\operatorname{Mor}_F(K, \overline{F})| < [K : F]$ contradicting the very first part of the lemma (already prove above as well). □